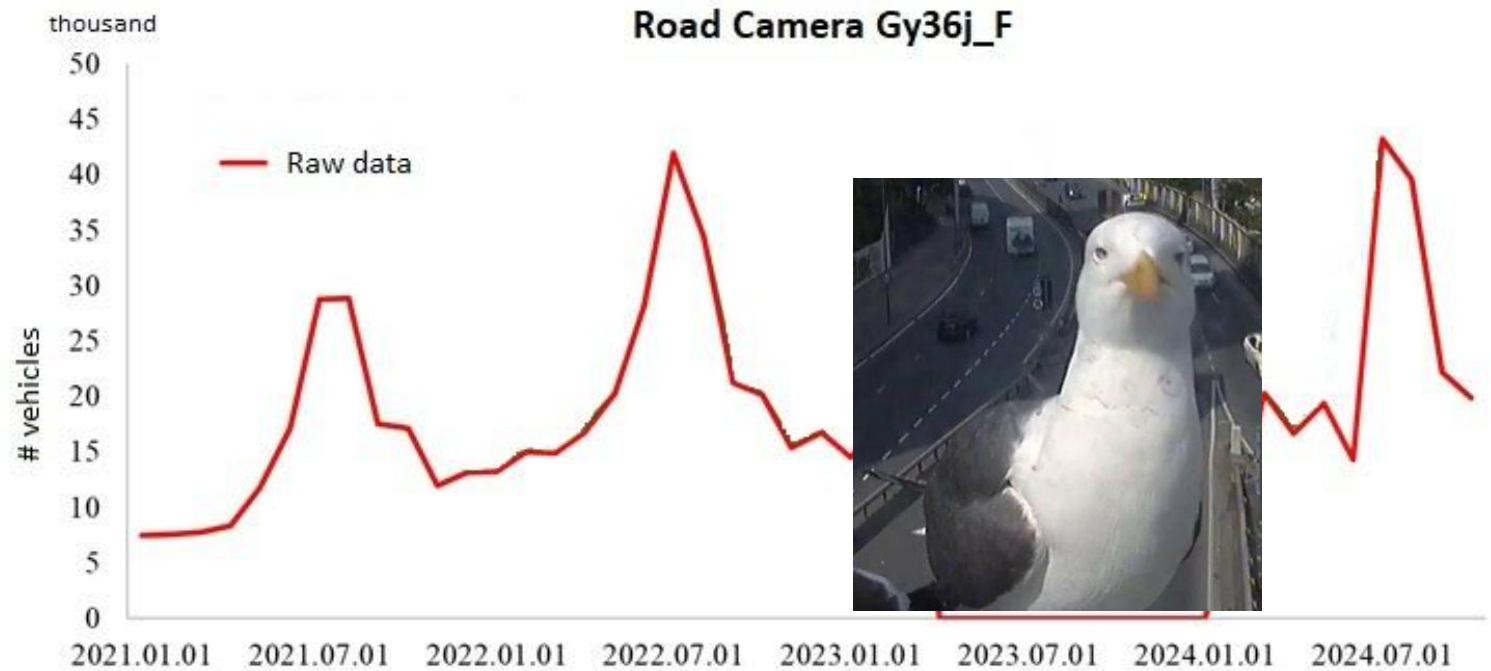


Machine Learning Border Crossing Traffic

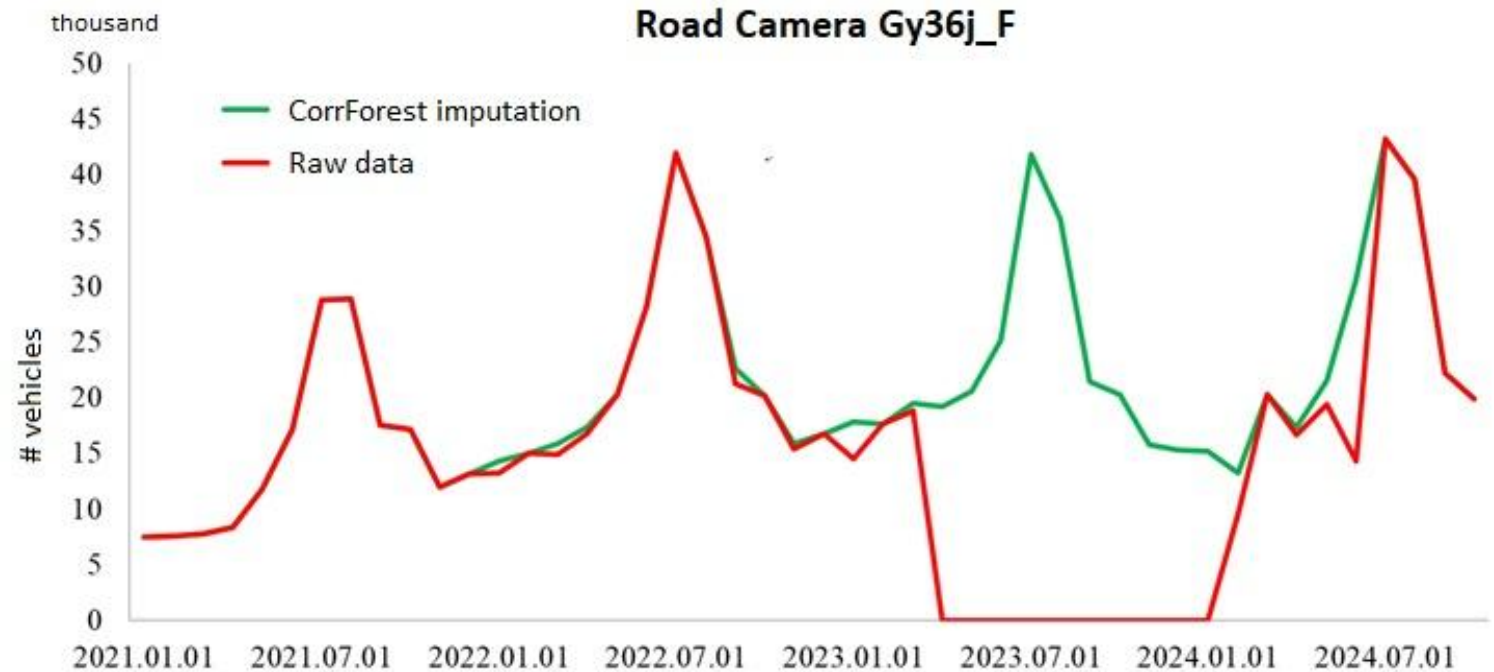
- ❑ aggregated monthly road camera data
- ❑ correlation analysis
+
- ❑ ML random forest
- CorrForest



$$I_G(p) = \sum_{i=1}^J \left(p_i \sum_{k \neq i} p_k \right) = \sum_{i=1}^J p_i (1 - p_i) = \sum_{i=1}^J (p_i - p_i^2) = \sum_{i=1}^J p_i - \sum_{i=1}^J p_i^2 = 1 - \sum_{i=1}^J p_i^2. \quad \Delta\sigma^2 = \sigma_p^2 - \left(\frac{n_l}{n_t} \sigma_l^2 + \frac{n_r}{n_t} \sigma_r^2 \right)$$

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Machine Learning Border Crossing Traffic

Traffic at border crossing points is strongly correlated with industrial production. And we want to make use of this while estimating. The traffic itself is measured by using road cameras.

The problem arises when a camera temporarily goes offline for some reason.

We addressed this issue using correlation analysis, taking advantage of the fact that the network of cameras and roads forms a well-modelable system.

By combining this with a random forest–based method, we developed our own algorithm, called CorrForest, which can accurately impute missing data. As you can see it grabs the underlying pattern satisfyingly.